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Uniformity of Powder Die Filling Using a Feed Shoe: A Review

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Uniformity of Powder Die Filling Using a Feed Shoe: A Review

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The process of die filling for achieving a uniform fill is a critical unit operation that is far from being sufficiently understood. Factors influencing powder flow from a feed shoe and powder deposition into a die are reviewed in this article. The factors include feed shoe speed, feed shoe holding time over the die, particle shape and size, powder size distribution, die shape and size, die configuration, filling intensity, filling direction, and the height of powder in the feed shoe. Devices suitable for evaluating the quality of powder deposition in dies need to be more: (1) specific to simulate the filling process, (2) versatile to encompass different filling parameters, and (3) able to provide quantitative evaluation. Recommendations to improve die deposition and density uniformity are given with respect to powder characteristics and die and feed shoe properties. New devices using load cells, pressure sensors, or tactile sensors to investigate powder deposition are important for further the investigation of the deposition process and spatial density uniformity. These primary measurement devices have the potential to bridge the knowledge gap and enable the development of discrete and/or continuum-concept computational models for the dynamic die-fill process.

Keywords die filling, feed shoe filling, powder flow, uniform fill

Introduction

A uniformly distributed bulk density is always desirable for powder filled into a die to make powder compacts. Inhomogeneity may cause many quality problems for the product, such as distortion, cracking, low strength, and shrinkage. In addition to physical properties, other properties, such as magnetic and chemical properties, of the product may be compromised. Higher fill density is also generally desirable. Pressing problems are reduced when fill density is high (Reed, 1995).

In the first part of this article, the filling process is discussed. The second part focuses on powder flow from feed shoe. Deposition of powder into a die or dies is discussed in the third part. Then, the state of the art in research on die filling is

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summarized. In the last part, recommendations are proposed to improve the uniformity of powder deposition and to further the investigation of this problem.

General Review of Die Filling by Feed Shoe

Gravity filling and suction filling are two common die-filling methods using a feed shoe. Powder flows into a die under gravity in gravity filling, while powder is "sucked" into the die by downward motion of movable punches in suction filling (Wu & Cocks, 2004). This article discusses only the filling process using the gravity filling.

Cocks et al. (2001) and Wu et al. (2003a) noticed that during the forward stroke of the feed shoe movement, for a not too high powder level in the feed shoe, the front powder adopted a nose-shaped profile due to inertia (Figure 1). They referred to powder flow from the tip as "nose flow" and that from the bottom as "bulk flow." Once the tip of the nose passed the cavity, the overall flow rate decreased noticeably. Under all conditions examined (0.1–100 m/s^2 acceleration and 0.05–1 m/s feed shoe speed), nose flow dominated the filling process. When the tip of the nose passed the cavity, the cavity was fully covered and the air in the cavity could escape only by permeating through the powder. This generated a pressure that opposed the powder flow, especially for fine and low-density particles. Before the tip of the nose passed the cavity, air could escape from the gap ahead of the nose. Air trapped in the die might result in low-density areas. Particles on the surface of the nose rolled down into the cavity with relative ease. However, it was difficult for particles to detach from the bulk of the powder mass because of interlocking with each other, which was reinforced when the feed shoe accelerated. On the other hand, the benefit of nose flow needs to be balanced with bulk flow, as only a small portion of the die opening is used.

Deposition uniformity is related to filling conditions (such as filling method and rate, feed shoe speed and acceleration, feed shoe vibration, additional filling cycles, powder level in feed shoe, and powder level in hopper), powder feed system design (such as hopper size and slope angle, tube size, and feed shoe size and geometry), die geometric characteristics and vibration, and powder characteristics (such as particle size and shape, size distribution, bulk density, and flowability) (Bruff & Jenike, 1967; Dhanoa & Puri, 1998; Sawayama & Seki, 1999; Mittal & Puri, 2001; Hjortsberg & Bergquist, 2002; Guillot et al., 2002). Ambient temperature and humidity are also known to affect die filling.

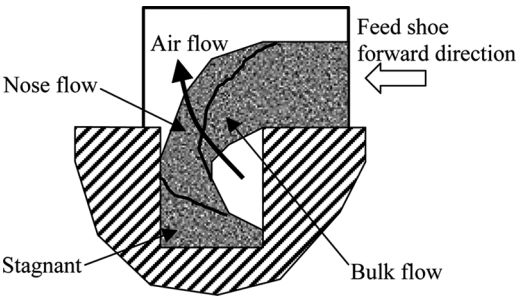


Figure 1. General features of die-filling process (Cocks et al., 2001).

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Flow of Powder from Feed Shoe

Flow of powder from feed shoe requires consistent and rapid flow. To achieve this goal, powder flow properties and powder discharge rate through an orifice are all of prime concern.

Powder Flow Properties

Dense, spherical, and large ($>20\,\mu m$ and below a critical range) particles with a smooth and nonstick surface are preferred for good flowability (Abdullah & Geldart, 1999). For instance, 5% or more fines ($<20\,\mu m$) may stop flow altogether. Extremely coarse granules may impede flow (through bridging) and the achievement of a uniform fill density (Reed, 1995). Powders having better flowability are likely to decrease part-to-part weight variation (Suzuki et al., 1997).

Fine and dry powders have lower values of angle of internal friction, while coarse and wet solids and cohesive powders have higher values (Fayed & Otten, 1997). Angle of internal friction can be measured by many apparatuses, such as the Jenike shear cell (Jenike, 1961), ring shear cell (Walker, 1967), and true triaxial tester (Kamath & Puri, 1990). The problem is that the information gained from these testers is accurate only for powder in steady-state flow (Wu & Cocks, 2004).

There are at least 16 different measuring techniques or equipment for evaluation of bulk solid flowability (Schulze, 1998). The Hall Flowmeter is a standard test device for metal powders (ASTM, 2004b), but the flow rate obtained from it cannot be used to predict either the limiting feed shoe speed or the flow rate through a feed hopper (Wu & Cocks, 2004). The test results are also affected by several interrelated powder attributes, such as particle shape, particle size, and the permeability of the powder (Bell, 1993).

Compared with the Hall Flowmeter (ASTM, 2004a), the Arnold Meter is more sensitive to small changes in apparent density and gives results more representative of flow in industrial die-filling operations (Peterson & Small, 1994). A few other devices were created to evaluate powder flow or deposition behavior (Larsson & Vidarsson, 2000; Freeman, 2001), but either the connection of the test data with actual application is not straightforward or they provide only limited information.

Rice and Tengzelius (1986) indicated that it is unlikely to have a single apparatus or method that is suitable to test various powders' flowability or die-filling characteristics. Wu et al. (2003b) introduced the concept of critical velocity, above which complete filling of a standard rectangular die by a feed shoe cannot be achieved.

Ways to improve flowability and, consequently, the rate of powder deposition include making dense, spherical, smooth, and reasonably large particles and reducing the moisture content. Current apparatuses to measure powder flowability are not suitable for evaluating the powder deposition process. During flow, the powder starts from stationary condition to flow state and exits from an opening and experiences conditions that are significantly different during deposition when a moving powder is brought to stationary conditions.

Discharge Rate through an Orifice

As few published studies are available to predict powder flow rate from a moving feed shoe, hopper flow is reviewed to provide background knowledge to understand powder flow from a feed shoe and identify major factors affecting the process.

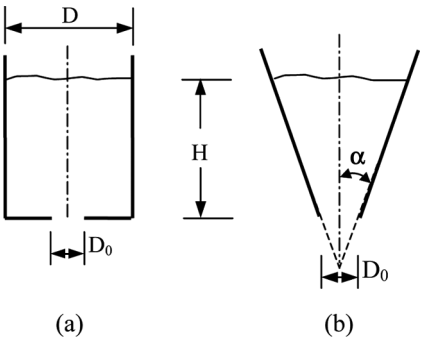


Figure 2. Notation in (a) cylindrical and (b) conical hoppers (Nedderman et al., 1982).

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Nedderman et al. (1982) reviewed discharge rate of granular materials from typical cylindrical and conical hoppers (Figure 2). Mass discharge rate (W) is independent of granular material fill height (H) if H is larger than 2.5 D or D (hopper diameter) (Hinchley, 1926; Ketchum, 1929; Brown & Richards, 1959), or even H is greater than the orifice diameter D_0 (Rose & Tanaka, 1959; Myers & Sellers, 1978).

Ketchum (1929), Brown and Richards (1960), and Franklin and Johanson (1955) reported that W is constant provided $D > 2.5D_0$ or $D\text{-}D_0 > 30d$, where d is the particle diameter. As D/D_0 decreased, W increased. In the limit as $D \rightarrow D_0$, the whole mass of material accelerates indefinitely under gravity.

Beverloo et al. (1961) proposed the following equation:

$$W = C\rho\sqrt{g}(D_0 - kd)^{\frac{5}{2}} \tag{1}$$

where W is mass discharge rate, C is Beverloo coefficient, ρ is bulk density, g is gravitational acceleration, D_0 is orifice diameter, k is dimensionless constant, and d is particle diameter.

Equation (1) breaks down for fine powders ($d < 500$ or 600 μm) (Nedderman et al., 1982; Verghese & Nedderman, 1995). This is because the powder approaching the orifice is subjected to an adverse pressure gradient (Crewdson et al., 1977).

Verghese and Nedderman (1995) adjusted Equation (1) by a coefficient, which implied that the mass flow rate is zero for particles less than 120 μm. Equation (1) should not be used if $D_0 < 6d$, as the flow is intermittent and irreproducible (Nedderman et al., 1982).

Flow rate through an orifice was also investigated by many other researchers, such as Wieghardt (1952), Brown and Richards (1965), Kotchanova (1970), Mroz and Szymanski (1971), Chang et al. (1984, 1991), and Zhang and Rudolph (1991). ASAE Standard D274.1 (ASAE, 2003) provides an equation to calculate the flow rate of grain and oil seeds through an orifice for mass flow situations.

Deposition of Powder into a Die or Dies

Uniform fill density distribution, less segregation, minimal air entrapped, and high density (for some cases) are desirable for powder deposition into a die or dies. Powder properties, feed shoe features, and die characteristics are all factors affecting powder deposition.

Improved powder flowability can improve fill density distribution and reduce size variation of green compacts. The drawback is that the particles are more inclined to segregate (Larsson & Vidarsson, 2000).

Filling rate decreases with the decreasing average grain size and the increasing irregularity of the particle shape because of the dependence of cohesion upon particle size and morphology (Wu et al., 2003a). Another reason is that the effect of airflow from the die becomes more significant for smaller and more irregular particles.

Particle Packing

Variables that may affect particle packing are listed in Table 1 (Gray, 1968). It is possible to adjust the particles to attain high packing densities or to attain specific compact properties (German, 1989). Powder characteristics that retard flow may hinder packing and produce density nonuniformity (Reed, 1995).

Friction, adhesion, and other surface forces become increasingly important with decreasing particle size, and this may promote bridging and arching and increase porosity (Gray, 1968). Sohn and Moreland (1968) found higher packing density for extended size distributions. Wakeman (1975) indicated that the density of packing of spheres in a loosely packed arrangement is not as critically dependent on the particle size range as in a densely packed arrangement.

Particle angularity increases porosity by raising interparticle friction and promoting bridging (Oman & Watson, 1944; Mittal et al., 2001b). Voidage decreases when round particles are added to a mixture of angular and round particles (Shergold, 1953). However, Wakeman (1975) reported that denser packing is possible with angular particles than with spherical particles upon vibration. According

Table 1. Variables controlling the packing of particles as summarized by Gray (1968)

Variable		Experimental observation	Theoretical basis
Particle	Shape	*	**
	Absolute size	*	*
	Size distribution	**	**
	Mass	*	*
	Elasticity	**	*
	Resilience	**	*
	Surface properties	*	*
Container	Shape	*	*
	Size	**	*
	Elasticity	●	●
	Surface properties	●	●
Deposition	Intensity of deposition	**	*
	Velocity of depositing particles	**	*
	Method	*	●
Treatment after deposition	Vibratory compaction	**	*
	Pressure compaction	**	*

● solid circles; *qualitative evidence; **quantitative evidence.

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to Reed (1995), the vibrated packing density of coarse angular particles of uniform size is 50–60% versus 65% for smooth spheres. A higher porosity is generally observed when the aspect ratio of the particles is high and when a higher content of fibers is added into isometric particles. Fill density is often increased by vibration, but vibration at too high a frequency may result in fluidization and low packing density (Rice & Tengzelius, 1986).

Gray (1959) examined two kinds of packings, pouring particles into a container with zero impact velocity and with high impact velocity. The former resulted in a high porosity at the wall, a slightly lower porosity in the second layer from the wall, and a more or less constant value throughout the rest of the container. The latter resulted in closer packing at the wall and also closer packing in the succeeding layers. Until four layers from the wall, the influence of the wall was no longer apparent, and porosity increased to the level of an infinitely extensive bed.

Benenati and Brosilow (1962) found that a minimum voidage occurs at a distance of about half a particle diameter from the wall, i.e., at the center of the first layer of spheres. The cyclic variation in voidage extends to a distance of 4.5 to 5 particle diameters before it dissipates. Ridgway and Tarbuck (1966) indicated that mixing spheres of different sizes causes a more rapid approach to randomness than monosize spheres.

Gray (1968) pointed out that almost all packings comprise a central core undisturbed by the walls. For a ratio of container size to particle size less than 50, the overall porosity is affected by the wall, and it may be important to consider the detailed variation in voidage near the wall. Wadsworth (1960) proposed that in random packing porosity is a function only of the ratio of container to sphere diameter, and it is fairly insensitive to the method of stacking.

Graton and Fraser (1935) suggested that packing can be influenced by filling intensity and particles' speed and direction. A too low velocity leads to a high porosity (Kolbuszewski, 1950). The overall porosity decreases with increasing impact velocity, and there is a critical filling intensity that produces the closest packing (Macrae & Gray, 1961).

Segregation takes place during deposition. Tang and Puri (2004) summarized universal methods to minimize segregation, including improvement of material properties (i.e., narrowing size distribution, reducing particle size, and avoiding irregular-shaped particles), proper selection of handling equipment and parameters (i.e., lowering free-fall height and employing mass flow bins), and environmental control (i.e., minimizing vibration and maintaining humidity). Unique methods to minimize segregation in the deposition process are to prevent heap from forming, or reduce the heap size and free-fall height. Other possible ways include granulating the fines and controlling air currents.

Size segregation is probably the most common segregation in die filling, and it makes pockets of near single-sized particles, a situation characterized by high porosity (Gray, 1968). One of the ways of reducing segregation is to deposit at high intensity and from low heights (low impact velocity), conditions that form a bed of high porosity (Richards, 1966).

Lawrence and Beddow (1969) poured particles into a die via a cone and found a fines-rich inner mound on the bottom of the die and an excess of coarse particles in the periphery. Reducing the size of the fines increased segregation up to the point where the relative size difference (difference between coarse and fine divided by coarse) is near 0.7. For fines smaller than this, flow is restricted and segregation is

reduced. Segregation increases to a maximum at 15–30% fines percentage, and it approaches zero at 60% fines percentage. Lawrence and Beddow (1968) found that vibration can reduce radial segregation by leveling off the inner mound of fines. The effect of vibration on vertical segregation depends upon vibration amplitude and frequency.

Size segregation is less of a problem for fine particles ($< 100\,\mu m$) due to high interparticle friction. Segregation increases with mean particle size from 100 to 500 μm. Above approximately 500 μm, no further increase was observed (Williams, 1976).

Bridging of particles and agglomerates is significant when the die size is small ($< 10\,d$) and particle surface and interface are rough. Lubrication of all surfaces and vibration may facilitate a denser packing, but they may also promote agglomeration and segregation (Reed, 1995).

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Effect of Feed Shoe Features

Kondoh and Takemoto (1996) indicated that fill density slightly increases with increasing feed shoe speed (from 10 to 60 mm/s). At 10 mm/s speed, particles roll down the slope and gradually fill the die. At 60 mm/s speed, particles quickly fall into the die. The air in the cavity is intensely expelled out, and the particles dropping down later are blown away to the front. Particle deposition in the die can be divided into three regions (Figure 3). The top region, i.e., rolling region, is dragged by the feed shoe during the backward stroke and moves horizontally to the right side. The sliding region slides diagonally with the backward movement of the feed shoe. In the center region of the die, the moving distances are the largest. The bottom region, i.e., non-sliding region, does not move.

Hjortsberg and Bergquist (2002) also stated that fill density at higher feed shoe speed (1400 mm/s) is larger than at lower speed (600 mm/s). Wu and Cocks (2004) indicated that at higher feed shoe speed the powder stream has higher kinetic energy, and the oncoming particles collapse some bridges in the die and promote higher fill density, especially for a large die. But Larsson and Vidarsson (2000) pointed out that with increasing feed shoe speed from 30 to 200 mm/s, cavities tended to be only partially filled, especially for small cavities. Rice and Tengzelius (1986) observed that the slower the speed, the greater the fill density. Increasing the number of strokes of the feed shoe is of significance at fast feed shoe speed, which suggests that holding time of the feed shoe over the die is an important factor.

Wu et al. (2003a) reported that die filling rate slightly decreases with increasing feed shoe speed, due to increasingly limited nose flow and increasing powder

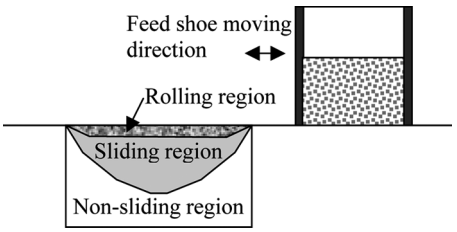


Figure 3. Particles moving during pullback of feed shoe (Kondoh & Takemoto, 1996).

cohesion because of a longer period of acceleration. High feed shoe speeds tend to result in incomplete filling. Wu et al. (2002) found higher filling rate for a stationary feed shoe, though only bulk flow occurs, than for a moving one. They attributed this to increased cohesion due to the initial acceleration for a moving feed shoe. For the stationary feed shoe, the orifice available for powder flow is always the die opening; while for the moving feed shoe, the orifice varies from zero to the full die opening.

Sawayama and Seki (1999) developed an air-replacement feed shoe, an ordinary square feed shoe divided in half diagonally by a pipe to pass air from the bottom of the feed shoe. The air-replacement feed shoe can improve filling speed and parts weight variation. An aeration filling method was proposed for uniform fill (Kondoh & Takemoto, 1996; Urata et al., 1998). A few small pipes were installed at the bottom of the feed shoe to carry the gas blown into the powder via tiny holes on the pipes. This reduced friction between particles and reduced size segregation. It was reported to be especially effective for thin and complex-shaped cavities for decreasing weight variation and improving dimensional tolerance of powder products. Similar feed shoes were reported by Zahrah et al. (2001), Guillot et al. (2002), and Hashimoto et al. (2002). The downside of these new feed shoes is that they make the feed shoe more complicated and may promote segregation.

Haskins and Jandeska (1998) proposed that the shoe motion exerts an angled downward force due to static head pressure of powder in the feed shoe and the horizontal velocity vector of the feed shoe. The resultant downward force appears to be better transferred through the packed bed of low-friction powder.

Sawayama and Seki (1999) pointed out that shaking the feed shoe increases fill density, but holding (i.e., pausing) the feed shoe over the cavity had no effect. For larger die cavities, neither shaking nor holding helped. But Hjortsberg and Bergquist (2002) found that both holding and shaking increased fill density. Hjortsberg (2000) obtained similar results for shaking, but the variation of fill density within each die increased as well. Guillot et al. (2002) showed that higher and relatively evenly distributed bulk density is achieved by shaking.

Wu and Cocks (2004) confirmed that fill density increases with increasing shaking number. They attributed this to the shear zone between the powder inside the feed shoe and the powder in the die.

Effect of Die Characteristics

Fill density decreases with decreasing die opening (Rice & Tengzelius, 1986; Bocchini, 1987). Rice and Tengzelius (1986) suggested the reason is that particle rearrangement is less easily satisfied in smaller dies. As particles become large relative to the die, the influence of the die wall becomes important (Bocchini, 1987). Larsson and Vidarsson (2000) further pointed out that improving powder flowability reduces the decrease in fill density due to small die dimensions. Readey and Mahoney (1995) indicated that fill density also decreases with decreasing die aspect ratio (the ratio of compact thickness to die diameter).

Fill density distribution is also affected by die shape and size (Dhanoa & Puri, 1997; Mittal et al., 2001a; Mittal & Puri, 2001). Higher mass distribution was observed in the relatively wide middle leg of an E-shaped die. For a toroid die, the powder mass tends to concentrate in the direction parallel to the direction of the feed shoe.

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Incomplete fill is often observed for narrow or small dies. The kinematics of the filling process is more complex for dies with complex features, such as steps (Guillot et al., 2002; Wu et al., 2003a). Usually the deeper the die is, the lower the fill density (Guillot et al., 2002). Density varies little on the left and right sides of the feed shoe moving direction, but the density of the pullback end side tends to be higher than that of the advancing end side (Urata et al., 1998).

When the feed shoe is in the direction perpendicular to a rectangular cavity, the fill process is more difficult, especially for narrow cavities. For filling in parallel direction, the cavity remains open during most of the time while the feed shoe is moving across, thus allowing air to escape from the cavity (Vidarsson & Arvidsson, 2000).

Decreasing die size may reduce segregation. The reason is that the die is filled more rapidly, which reduces the time for segregation (Lawrence & Beddow, 1969).

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Effect of Selected Process-Related Factors

Increasing filling rate decreases fill density, but there is a limit beyond which the density will not decrease with increasing filling rate (Faikin et al., 1976). The reason is that higher filling rate reduces the time for particles to rearrange and for the trapped air to escape. Applying vibration to the die may help counteract this negative effect. Segregation may be reduced by increasing the filling rate and increasing the height from where the powder falls into the die (Lawrence & Beddow, 1969).

Fill density distribution is also affected by die filling rate (Dhanoa & Puri; 1998). Mittal et al. (2001a) indicated that feed shoe moving direction affects the powder deposition behavior in complex-shaped and/or multiple-corner dies such as an E-shape.

Sawayama and Seki (1999) reported that high powder level in the feed shoe prevents the die from being fully filled at high feed shoe speed. Effects of powder level on displacement of the air in the cavity, interparticle friction, and/or adhesion were considered as possible reasons. But fill density is lower and varies more for lower powder level in the feed shoe. Rice and Tengzelius (1986) found that higher powder level in the feed shoe may lead to higher fill density. However, Hjortsberg and Bergquist (2002) indicated that fill density is not sensitive to the amount of powder in the feed shoe.

Completely filling the feed shoe may reduce powder flow rate. Flow rate can be increased by creating multiple noses, which can be achieved by dividing the feed shoe into a number of compartments using transversal separating walls (Wu et al., 2003b).

Tests of Powder Die Deposition

Differences in flow characteristics and densities can be detected by computed tomography (CT) analysis. Absolute density is determined within 1% to 2% accuracy, and relative density difference may be more precisely estimated (Haskins & Jandeska, 1998). Farber et al. (2003) also used CT technology to study the porosity and morphology of granules. However, the costs involved are very high, and it is simply impossible to test certain powders that would cause high attenuation of incident X-rays so as to render them undetectable by a CT scanner. Furthermore, only final density or mass values can be collected by CT technology, and no real-time deposition

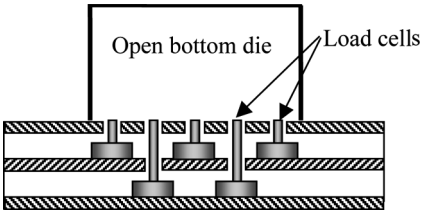


Figure 4. Schematic of mass deposition tester (not to scale) (Dhanoa & Puri, 1998).

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profile can be obtained by a CT scanner, partly due to its relatively slow inspection rate (Copley et al., 1994).

Dhanoa and Puri (1998) developed a mass deposition tester (MDT) to acquire real-time mass increase profile for a die-filling process using load cells (Figure 4). It consisted of a tiered arrangement of load cells and a thin plastic film used to isolate the fill powder from the load cells (Puri & Dhanoa, 2000). Then the second-generation mass deposition tester (MDT-II) with flexible design and fully automated features was developed (Xie & Puri, 2003, 2004). The computer-controlled MDT-II incorporates a pressure sensor strip to measure pressure or mass increase profile at multiple locations simultaneously. Results showed that the MDT-II was an effective and efficient device for evaluation of die deposition.

Demetry et al. (1998) employed a tactile sensor at the bottom of the die to detect mass distribution after die filling. Nonuniform mass distribution of the powder fill can be detected as gradients in transmitted pressure at the bottom of the die. Kondoh and Takemoto (1996) used a camera to record the filling process.

Hjortsberg and Bergquist (2002) built a device similar to industrial filling systems to study fill density and process variables. Density variation measurement was done by dividing the cavity into four sections through four plates inserted into the filled ring-shaped die. The powder of each section was then sucked out, separated by a filter, collected, and measured to get density values.

Related Computational Models

Three methods are widely used to analyze bin discharge or die filling, namely kinematic models, finite element simulation of continuum models, and discrete element simulation of particle assemblies (Sundaresan, 2001). Kinematic models are limited because they need the rate of discharge to be specified and the flow patterns inside the bin to be computed. Finite element (FE) codes cannot simulate the filling process in a realistic manner, so the initial state prior to discharge has to be specified. Challenges in the finite element approach include exit boundary conditions and reliable constitutive models. For the discrete element (DE) method, each particle is separately represented. The DE method is not feasible for a large number of particles with varied shapes and multiple normal and frictional contacts without using large fictitious particles to represent thousands of actual particles. Current DE programs are, therefore, restricted either to a small number of particles or to particles with very simple geometries, usually in two-dimensional space only (Holst et al., 1999).

There are a great number of FE and DE models in the literature, that are relevant to silo pressure or powder discharge from silos and the die-filling process, such as Zhang et al. (1986), Tejchman (1998), Kamath and Puri (1999), and Wu

et al. (2002). However, none of these is directly applicable to the evaluation and quantification of the dynamic die-filling process.

Summary of the State of the Art

Die deposition and the achievement of uniform fill, while seemingly simple, are deceptively complex processes that are far from being sufficiently understood. Very few, if any, current devices and methods assessing powder flowability can directly evaluate die deposition, due to the complexity of the process and device limitations. Devices suitable for this purpose need to be: (1) specific toward simulating the filling process, (2) versatile to encompass different filling parameters, and (3) able to provide quantitative evaluation. No literature has been located that studied the flow rate of powder from a moving feed shoe, except discharge rate of bulk solids from a bin. This fact makes research on feed shoe filling especially challenging and particularly rewarding.

Feed shoe speed, holding time over the die, and shaking of feed shoe have effects on die filling. New feed shoes with different features were created by many researchers to improve deposition. Particle packing and fill density and uniformity are also affected by particle shape and size, size distribution, die shape and size, die configuration, and other features, such as vibration. Other factors, such as filling intensity, filling direction, and the height of powder in the feed shoe, may also affect die filling. Different devices or methods, such as CT technology and high-speed imaging, have been used to investigate die deposition. The methods of using load cells, pressure sensors (Dhanoa & Puri, 1998; Xie & Puri, 2003, 2004), or tactile sensors (Demetry et al., 1998) provide quantitative information about the entire die fill process from empty to completely full state at reasonable cost; accordingly, these are the most practical and promising methods.

Recommendations

Making powder denser, increasing particle size, extending size distribution, decreasing particle irregularity, and decreasing moisture content are proven methods to enhance powder deposition, fill density, and fill uniformity. However, a balance needs to be maintained to minimize adverse effects such as segregation. Die shape and size (including sizes of different parts of the same die) need to be properly designed considering impact of powder deposition factors to improve fill uniformity. For dies of certain shapes, the feed shoe direction of approach can be critical. Based on the experience of previous researchers, it is necessary to use appropriate feed shoe speed while maintaining a certain powder level in the feed shoe.

New devices are especially important for further investigation of the deposition process and density uniformity distribution. Methods employing load cells or pressure sensors or tactile sensors are practical and promising. They are less expensive, can provide quantitative values, and record the entire deposition process. New features can be added, such as using lead (II) acetate trihydrate to label the particles of low atomic weight and make them visually traceable on CT scanners (Yang & Fu, 2004). Inclusion of radioactive materials into powder samples to monitor the behavior of the powder may also help. For new powder formulations, it is suggested to first focus on feed shoe-die shape related parameters such as die orientation with

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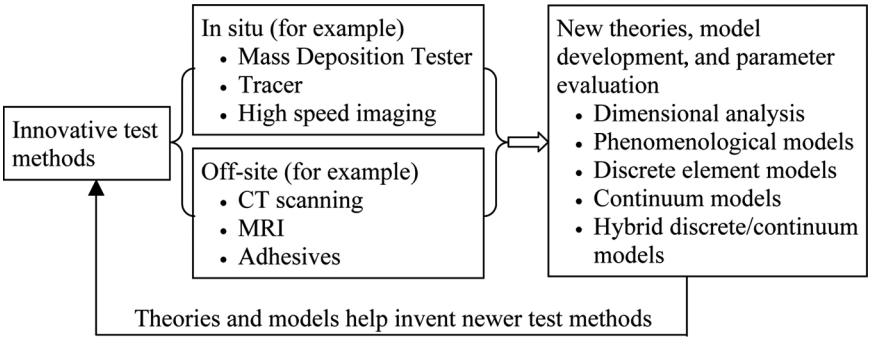


Figure 5. Strategies to improve the understanding of powder die filling.

respect to feed shoe, feed shoe shape, angle, acceleration, and speed values, and powder fill height in feed shoe.

One of the major obstacles for studying die filling is that physical dimensions of the dies are often too small (e.g., in the order of a few millimeters), and this hinders effective measurement and study of fill density distribution and the filling process. To overcome this, similarity theory can be used and large-scale models (devices) can be fabricated to study the problem.

By using new devices, innovative theories of powder deposition could be developed to advance the understanding of the die-filling process; the innovative theories in turn may promote development of a new generation of devices (Figure 5). The updated devices and novel theories, together with currently available theories, are expected to contribute to generating computational models, which is the ultimate goal. The computational models should incorporate major factors affecting the deposition process and mass distribution after die filling. The models are anticipated to enhance the understanding of the complex deposition process that will enable real-time monitoring, control, and evaluation of die fills.

Given the complexity of the powder deposition process, dimensional analysis can be used to develop preliminary models and discover physical laws for powder fill. Different models or modeling methods, such as discrete element models and finite element models, may be combined to simulate the filling process. Figure 5 summarizes some of the strategies or methods to advance the understanding of the process of powder die filling, including innovative test methods, new theories, and models. For example, a mass deposition tester can be used to collect data for the die-filling process. The data collected can be used to develop models and even expand current theories on die filling, and in turn, the new models and theories may contribute to inventions of even newer testing methods or devices.

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